# Project Coversheet

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| Full Name | Khalid Takar |
| Project Title (Example – Week1, Week2, Week3, Week 4) | Week 4 |

**Instructions:**

Students must download this cover sheet**,** use it as the first page of their project, and then save the entire document as a PDF before submission**.**

## Project Guidelines and Rules

### 1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

### 2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

### 3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.

- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

### 4. Plagiarism Policy

- Submit original work; no copy-pasting.
- Cite external material in a consistent format (e.g., APA, MLA).

### 5. Evaluation Criteria

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

### 6. Deadlines and Late Submissions

- Deadline: Submit on time; trainees who fail to submit the project will miss the “Certificate of Excellence”

### 7. Additional Resources

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

**YOU CAN START YOUR PROJECT FROM HERE**

# TechHub Sales Performance Analysis

## 1. Introduction

This report presents an analysis of TechHub's sales performance using transactional sales data, customer demographic data, and product information. The objective of the analysis is to gain insights into overall revenue performance, identify trends over time, and understand how revenue varies by product category, region, and customer demographics.

An interactive dashboard was created using Tableau Public to visualise these insights in a clear and accessible way.

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## 2. Data Overview

The analysis is based on three datasets:

- **Sales Data**: Order date, revenue, quantity, product category, and region
- **Customer Data**: Customer age group and gender
- **Product Data**: Product identifiers and categories

The datasets were cleaned and prepared in Tableau by ensuring correct data types, consistent categories, and appropriate aggregation levels for analysis.

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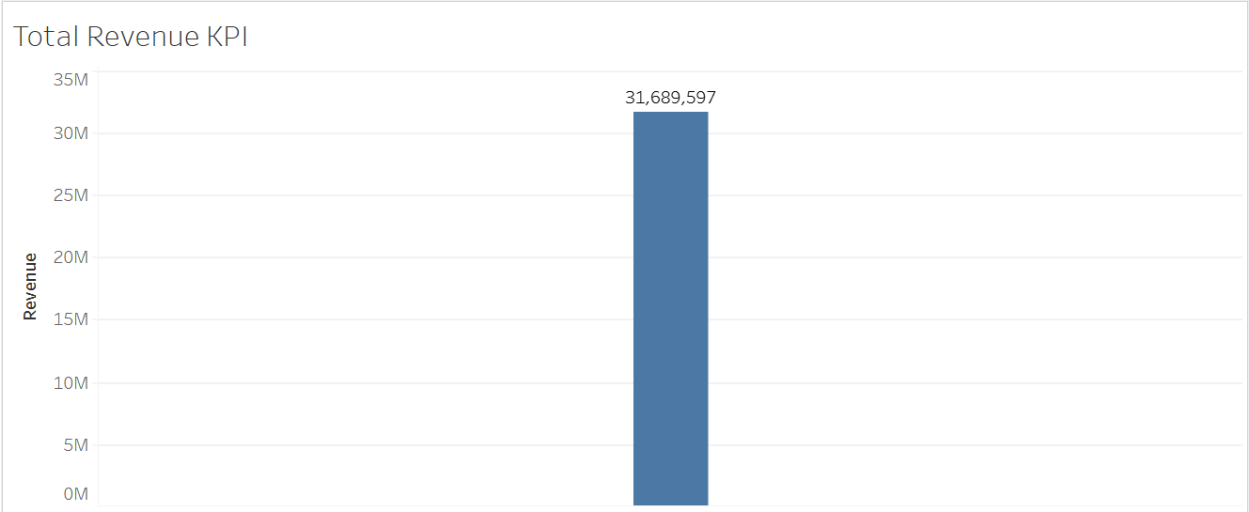
## 3. Total Revenue KPI

The dashboard includes a key performance indicator (KPI) showing total revenue across all recorded sales.

- **Total Revenue:** £31,689,597

This KPI provides a high-level overview of business performance and serves as a reference point for the other visualisations.




**[Figure 1: Total Revenue KPI]**

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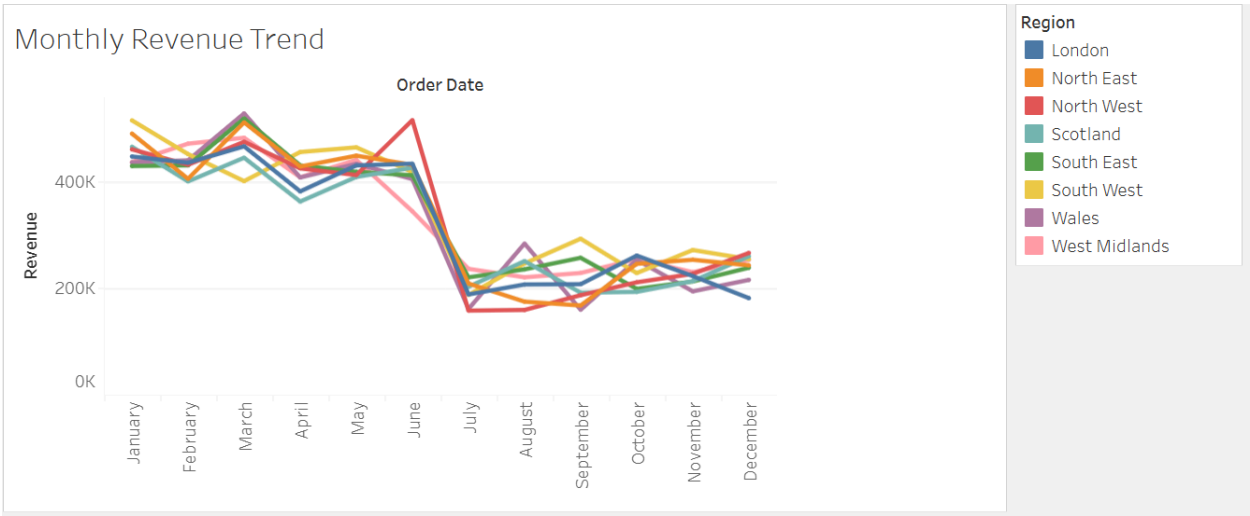
## 4. Monthly Revenue Trend

The monthly revenue trend visualisation shows how revenue changes over time across different regions.

Key insights:

- Revenue is relatively strong in the early months of the year.
- A noticeable decline occurs around mid-year, followed by stabilisation towards the end of the year.
- Regional trends follow similar patterns, suggesting seasonality rather than region-specific issues.

This trend analysis helps identify potential periods for targeted promotions or further investigation.




**[Figure 2: Monthly Revenue Trend by Region]**

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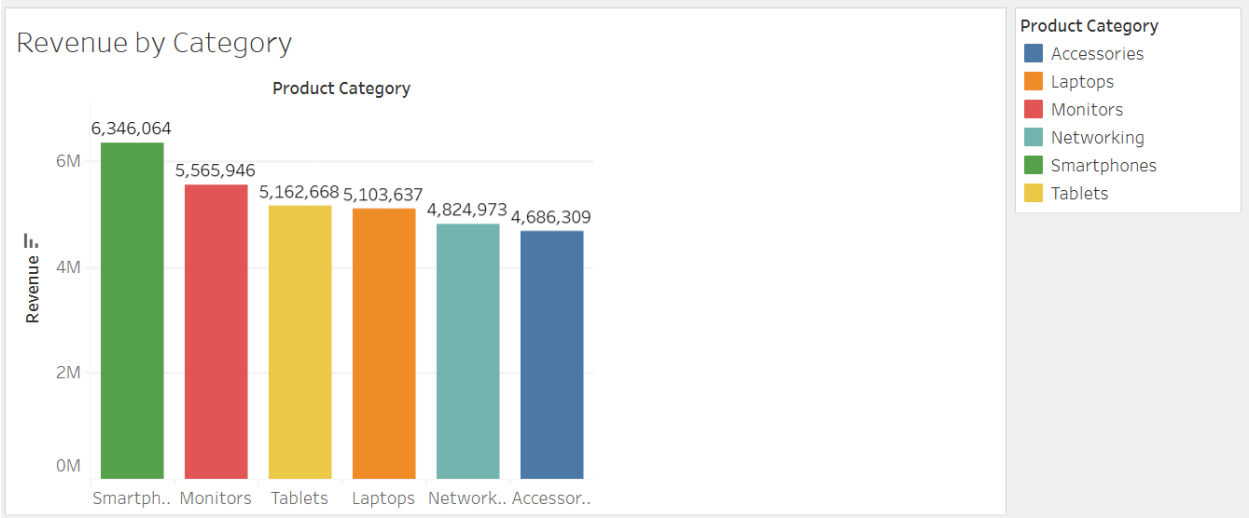
## 5. Revenue by Product Category

Revenue by product category highlights which product groups contribute most to overall sales.

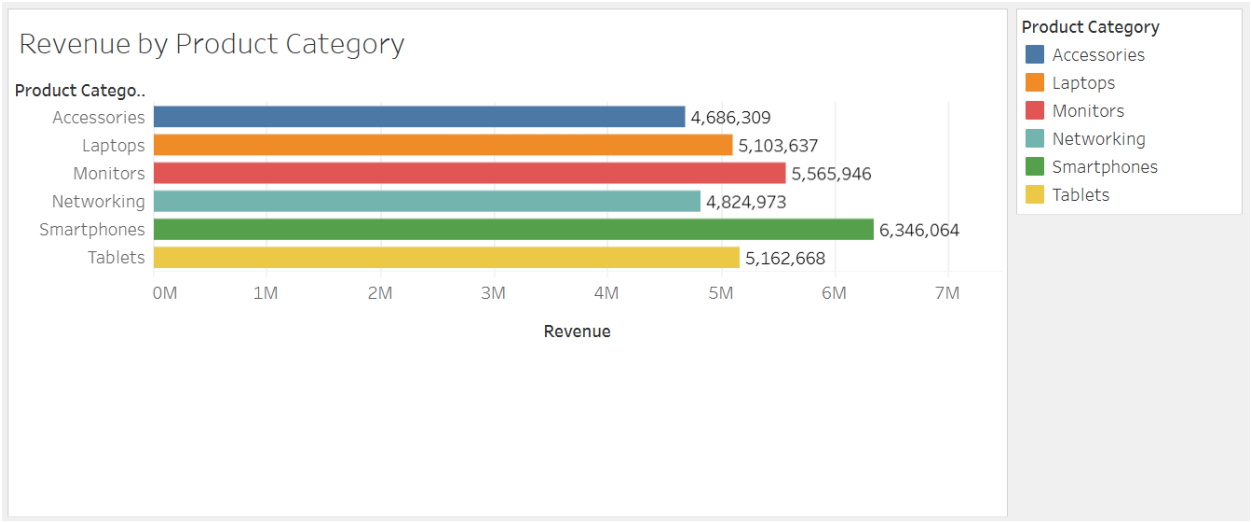
Key findings:

- **Smartphones** generate the highest revenue.
- **Monitors** and **Tablets** also perform strongly.
- **Accessories** generate the lowest revenue among categories.

These insights suggest that TechHub's core revenue is driven by major hardware products.




**[Figure 3: Revenue by Category]**




**[Figure 4: Revenue by Product Category]**

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## 6. Revenue by Region

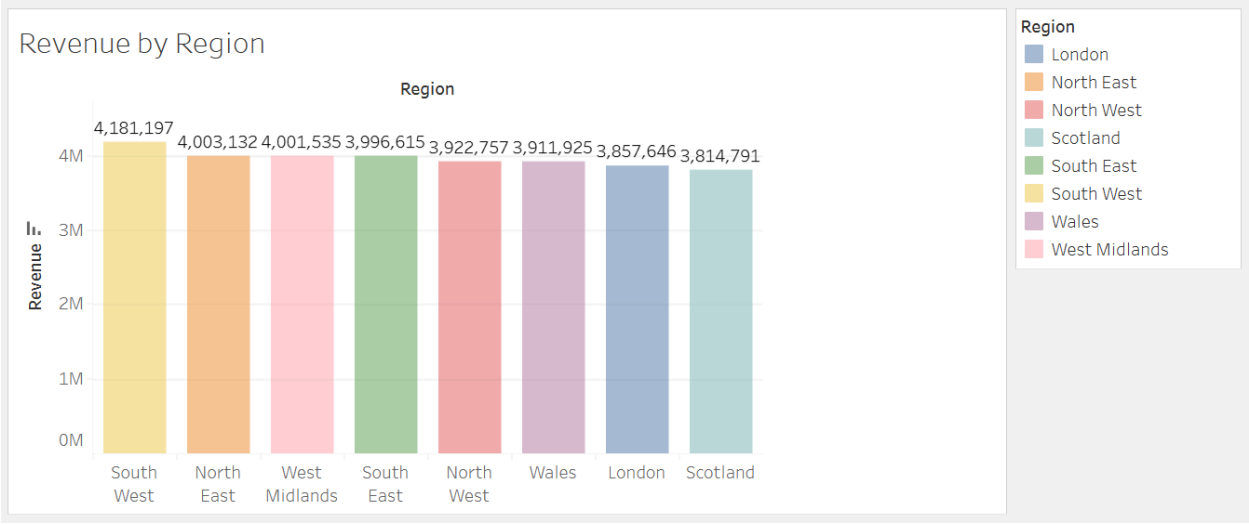
The regional revenue visualisation compares sales performance across different UK regions.

Key observations:

- Revenue is relatively evenly distributed across regions.
- The **South West** and **North East** show slightly higher revenue.

- **Scotland** and **London** generate comparatively lower revenue.

This indicates opportunities for targeted regional growth strategies.




**[Figure 5: Revenue by Region]**

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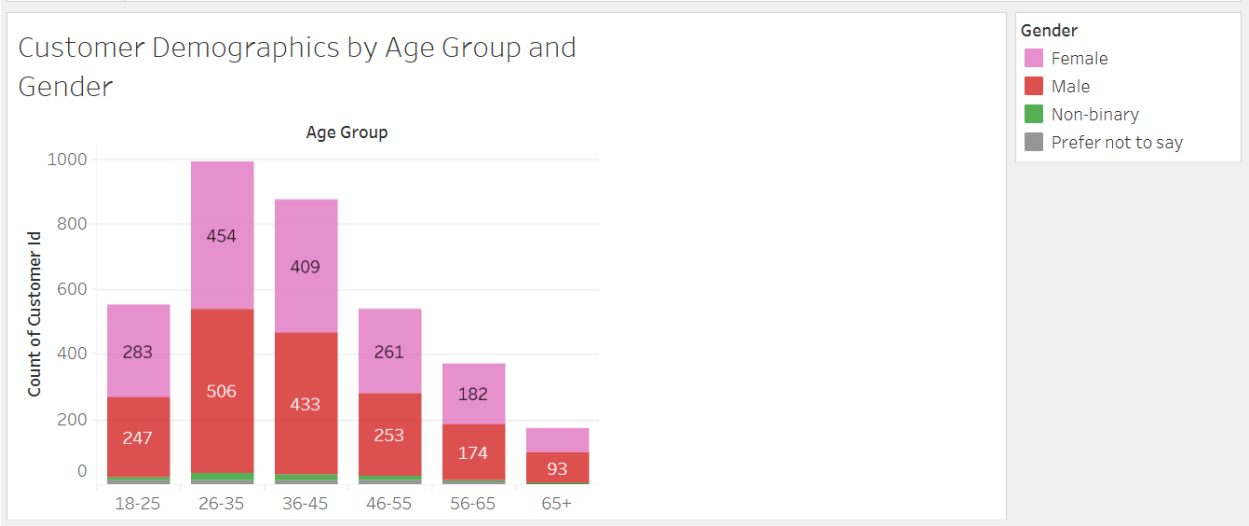
## 7. Customer Demographics Analysis

Customer demographics were analysed by age group and gender.

Key insights:

- The **26–35** and **36–45** age groups represent the largest customer segments.
- Female customers slightly outnumber male customers across most age groups.
- Customer counts decline in older age groups.

This suggests that TechHub's primary customer base consists of working-age adults.




**[Figure 6: Customer Demographics by Age Group and Gender]**

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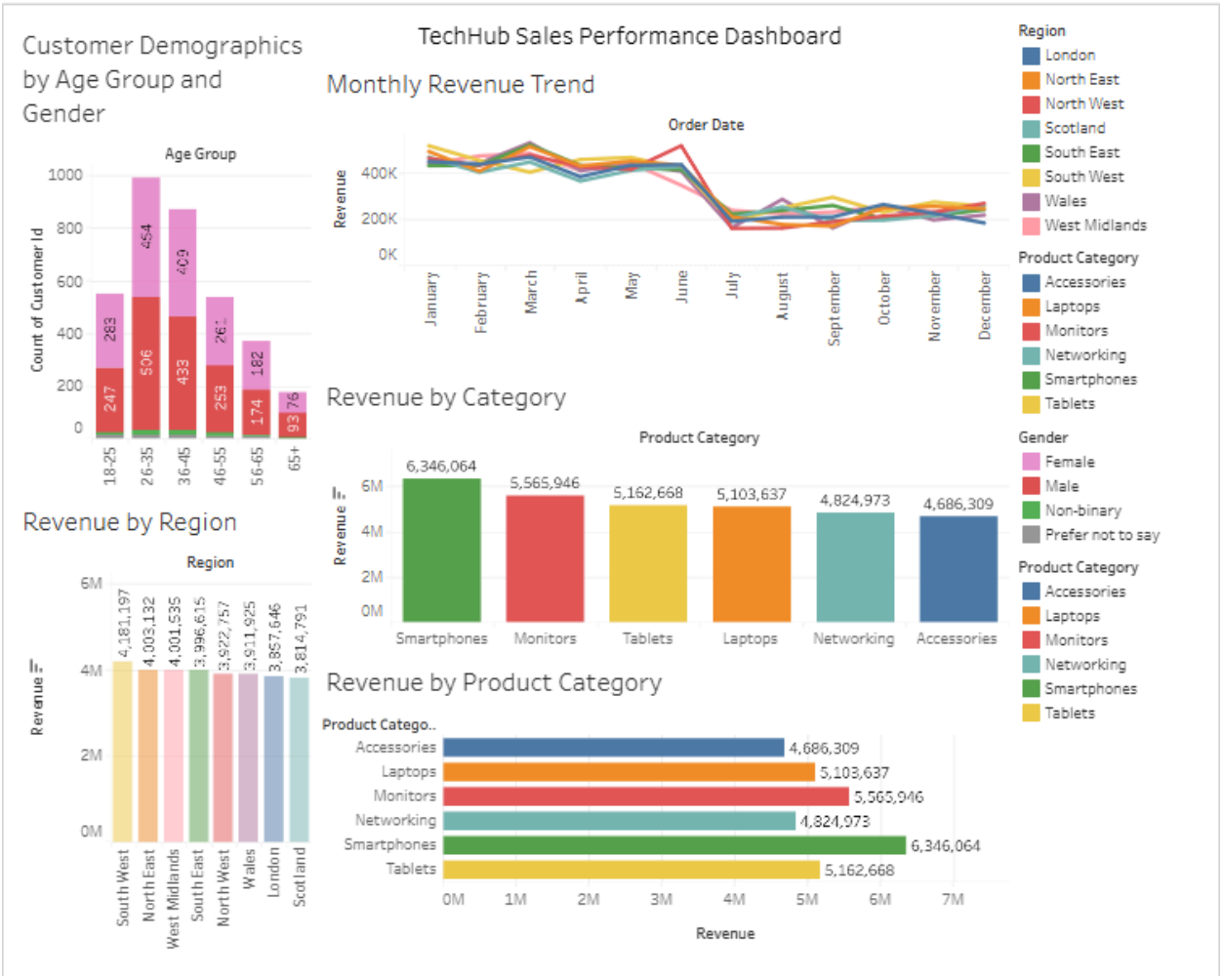
## 8. Dashboard Overview

All visualisations were combined into a single interactive dashboard to allow users to explore sales performance across multiple dimensions.

The dashboard enables:

- Monitoring overall revenue performance
- Exploring revenue trends over time
- Comparing product and regional performance
- Analysing customer demographics




**[Figure 7: TechHub Sales Performance Dashboard]**

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## 9. Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Focus on high-performing product categories such as Smartphones and Monitors.
2. Investigate the mid-year decline in revenue to identify underlying causes.
3. Target marketing efforts towards customers aged 26–45.
4. Develop region-specific strategies to improve performance in lower-revenue regions.

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## 10. Conclusion

This dashboard demonstrates how data visualisation can be used to transform raw sales data into actionable business insights. The findings provide a clear overview of TechHub's sales performance and support informed decision-making across products, regions, and customer segments.